

# SAFETY DATA SHEET

Revision Date 01-Sep-2023

Revision Number 3

## 1. Identification

**Product Name** 4-Fluorobenzonitrile

**Cat No. :** SB01283DA; SB01283R3; SB01283ZZ

**CAS No** 1194-02-1  
**Synonyms** p-Fluorobenzonitrile.; Benzonitrile, p-fluoro-; p-Cyanofluorobenzene

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

### Details of the supplier of the safety data sheet

**Company**

Fisher Scientific Company  
One Reagent Lane  
Fair Lawn, NJ 07410  
Tel: (201) 796-7100

Thermo Fisher Scientific  
One Reagent Lane  
Fair Lawn, NJ 07410

**Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
**Emergency Number US:**001-201-796-7100 / **Europe:** +32 14 57 52 99  
**CHEMTREC** Tel. No. **US:**001-800-424-9300 / **Europe:**001-703-527-3887

## 2. Hazard(s) identification

**Classification**

This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

|                                             |            |
|---------------------------------------------|------------|
| Acute oral toxicity                         | Category 4 |
| Acute dermal toxicity                       | Category 4 |
| Acute Inhalation Toxicity - Dusts and Mists | Category 4 |

**Label Elements****Signal Word**

Warning

**Hazard Statements**

Harmful if swallowed, in contact with skin or if inhaled

**Precautionary Statements****Prevention**

Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product  
Wear protective gloves/protective clothing/eye protection/face protection  
Avoid breathing dust/fume/gas/mist/vapors/spray  
Use only outdoors or in a well-ventilated area

**Inhalation**

IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing  
Call a POISON CENTER or doctor/physician if you feel unwell

**Skin**

IF ON SKIN: Wash with plenty of soap and water  
Call a POISON CENTER or doctor/physician if you feel unwell  
Wash contaminated clothing before reuse

**Ingestion**

IF SWALLOWED: Call a POISON CENTER or doctor/physician if you feel unwell  
Rinse mouth

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Hazards not otherwise classified (HNOC)**

None identified

### 3. Composition/Information on Ingredients

| Component               | CAS No    | Weight % |
|-------------------------|-----------|----------|
| Benzonitrile, 4-fluoro- | 1194-02-1 | 99       |

### 4. First-aid measures

|                                            |                                                                                                                   |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                      | If symptoms persist, call a physician.                                                                            |
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.   |
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician. |
| <b>Inhalation</b>                          | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.      |
| <b>Ingestion</b>                           | Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.             |
| <b>Most important symptoms and effects</b> | None reasonably foreseeable.                                                                                      |
| <b>Notes to Physician</b>                  | Treat symptomatically                                                                                             |

### 5. Fire-fighting measures

|                                         |                                                                                                                                |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Water spray. Carbon dioxide (CO <sub>2</sub> ). Dry chemical. Chemical foam. Water mist may be used to cool closed containers. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                                                                                       |
| <b>Flash Point</b>                      | 65 °C / 149 °F                                                                                                                 |
| <b>Method -</b>                         | No information available                                                                                                       |
| <b>Autoignition Temperature</b>         | No information available                                                                                                       |
| <b>Explosion Limits</b>                 |                                                                                                                                |
| <b>Upper</b>                            | No data available                                                                                                              |
| <b>Lower</b>                            | No data available                                                                                                              |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                                                                       |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                                                                       |

**Specific Hazards Arising from the Chemical**

Combustible material. Flammable. Containers may explode when heated.

**Hazardous Combustion Products**

Nitrogen oxides (NO<sub>x</sub>). Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Gaseous hydrogen fluoride (HF).

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**NFPA**

**Health**  
2

**Flammability**  
1

**Instability**  
0

**Physical hazards**  
N/A

## 6. Accidental release measures

|                                             |                                                                                                                                                                                          |
|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation. Remove all sources of ignition. Take precautionary measures against static discharges. |
| <b>Environmental Precautions</b>            | Should not be released into the environment.                                                                                                                                             |
| <b>Methods for Containment and Clean Up</b> | Sweep up and shovel into suitable containers for disposal. Keep in suitable, closed containers for disposal. Remove all sources of ignition.                                             |

## 7. Handling and storage

|                 |                                                                                                                                                                                                                                                                            |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Avoid dust formation. Keep away from open flames, hot surfaces and sources of ignition.                      |
| <b>Storage.</b> | Keep in a dry, cool and well-ventilated place. Keep container tightly closed. Keep away from heat, sparks and flame. Keep containers tightly closed in a dry, cool and well-ventilated place. Incompatible Materials. Strong oxidizing agents. Strong acids. Strong bases. |

## 8. Exposure controls / personal protection

**Exposure Guidelines**

| Component               | ACGIH TLV | OSHA PEL                           | NIOSH                      | Mexico OEL (TWA) |
|-------------------------|-----------|------------------------------------|----------------------------|------------------|
| Benzonitrile, 4-fluoro- |           | (Vacated) TWA: 5 mg/m <sup>3</sup> | IDLH: 25 mg/m <sup>3</sup> |                  |

Legend

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

**Engineering Measures** Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.

#### Personal Protective Equipment

**Eye/face Protection** Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.

**Skin and body protection** Wear appropriate protective gloves and clothing to prevent skin exposure.

**Respiratory Protection** Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

**Recommended Filter type:** Particulates filter conforming to EN 143.

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

### 9. Physical and chemical properties

|                                        |                              |
|----------------------------------------|------------------------------|
| Physical State                         | Solid                        |
| Appearance                             | White                        |
| Odor                                   | Odorless                     |
| Odor Threshold                         | No information available     |
| pH                                     | Not applicable               |
| Melting Point/Range                    | 33 - 36 °C / 91.4 - 96.8 °F  |
| Boiling Point/Range                    | 188 °C / 370.4 °F @ 750 mmHg |
| Flash Point                            | 65 °C / 149 °F               |
| Evaporation Rate                       | Not applicable               |
| Flammability (solid,gas)               | No information available     |
| Flammability or explosive limits       |                              |
| Upper                                  | No data available            |
| Lower                                  | No data available            |
| Vapor Pressure                         | No information available     |
| Vapor Density                          | Not applicable               |
| Specific Gravity                       | No information available     |
| Solubility                             | No information available     |
| Partition coefficient; n-octanol/water | No data available            |
| Autoignition Temperature               | No information available     |
| Decomposition Temperature              | No information available     |
| Viscosity                              | Not applicable               |
| Molecular Formula                      | C7 H4 F N                    |
| Molecular Weight                       | 121.11                       |

### 10. Stability and reactivity

|                                         |                                                                                                                |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <b>Reactive Hazard</b>                  | None known, based on information available                                                                     |
| <b>Stability</b>                        | Stable under normal conditions.                                                                                |
| <b>Conditions to Avoid</b>              | Keep away from open flames, hot surfaces and sources of ignition. Incompatible products.                       |
| <b>Incompatible Materials</b>           | Strong oxidizing agents, Strong acids, Strong bases                                                            |
| <b>Hazardous Decomposition Products</b> | Nitrogen oxides (NOx), Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> ), Gaseous hydrogen fluoride (HF) |

**Hazardous Polymerization** No information available.

**Hazardous Reactions** None under normal processing.

## 11. Toxicological information

### Acute Toxicity

#### Product Information

#### Component Information

**Toxicologically Synergistic Products** No information available

#### Delayed and immediate effects as well as chronic effects from short and long-term exposure

**Irritation** No information available

**Sensitization** No information available

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component               | CAS No    | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|-------------------------|-----------|------------|------------|------------|------------|------------|
| Benzonitrile, 4-fluoro- | 1194-02-1 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** None known

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** No information available

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Do not empty into drains.

| Component               | Freshwater Algae | Freshwater Fish | Microtox   | Water Flea               |
|-------------------------|------------------|-----------------|------------|--------------------------|
| Benzonitrile, 4-fluoro- | Not listed       | Not listed      | Not listed | 218 mg/l (Daphnia Magna) |

**Persistence and Degradability** Persistence is unlikely

**Bioaccumulation/ Accumulation** No information available.

**Mobility** No information available.

| Component               | log Pow |
|-------------------------|---------|
| Benzonitrile, 4-fluoro- | 1.74    |

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and

national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

### DOT

UN-No UN2811  
Hazard Class 6.1  
Packing Group III

### TDG

UN-No UN2811  
Hazard Class 6.1  
Packing Group III

### IATA

UN-No UN3439  
Proper Shipping Name NITRILES, TOXIC, SOLID, N.O.S.\*  
Hazard Class 6.1  
Packing Group III

### IMDG/IMO

UN-No UN3439  
Proper Shipping Name NITRILES, SOLID, TOXIC, N.O.S.  
Hazard Class 6.1  
Packing Group III

## 15. Regulatory information

### United States of America Inventory

| Component               | CAS No    | TSCA | TSCA Inventory notification - Active-Inactive | TSCA - EPA Regulatory Flags |
|-------------------------|-----------|------|-----------------------------------------------|-----------------------------|
| Benzonitrile, 4-fluoro- | 1194-02-1 | X    | ACTIVE                                        | -                           |

#### Legend:

TSCA US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

X - Listed

'-' - Not Listed

**TSCA - Per 40 CFR 751, Regulation of Certain Chemical Substances & Mixtures, Under TSCA Section 6(h) (PBT)**

Not applicable

**TSCA 12(b)** - Notices of Export

Not applicable

### International Inventories

China, X = listed, Australia, U.S.A. (TSCA), Canada (DSL/NDL), Europe (EINECS/ELINCS/NLP), Australia (AICS), Korea (KECL), China (IECSC), Japan (ENCS), Philippines (PICCS), Taiwan (TCSI), Japan (ISHL), New Zealand (NZIoC), Japan (ISHL).

| Component               | CAS No    | DSL | NDL | EINECS    | PICCS | ENCS | ISHL | AICS | IECSC | KECL |
|-------------------------|-----------|-----|-----|-----------|-------|------|------|------|-------|------|
| Benzonitrile, 4-fluoro- | 1194-02-1 | -   | X   | 214-784-9 | -     | -    | X    | X    | X     | -    |

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

### U.S. Federal Regulations

#### SARA 313

| Component               | CAS No    | Weight % | SARA 313 - Threshold Values % |
|-------------------------|-----------|----------|-------------------------------|
| Benzonitrile, 4-fluoro- | 1194-02-1 | 99       | 1.0                           |

**SARA 311/312 Hazard Categories** See section 2 for more information

#### CWA (Clean Water Act)

| Component | CWA - Hazardous | CWA - Reportable | CWA - Toxic Pollutants | CWA - Priority Pollutants |
|-----------|-----------------|------------------|------------------------|---------------------------|
|-----------|-----------------|------------------|------------------------|---------------------------|

|                         | Substances | Quantities |   |   |
|-------------------------|------------|------------|---|---|
| Benzonitrile, 4-fluoro- | -          | -          | X | X |

**Clean Air Act**

| Component               | HAPS Data | Class 1 Ozone Depletors | Class 2 Ozone Depletors |
|-------------------------|-----------|-------------------------|-------------------------|
| Benzonitrile, 4-fluoro- | X         |                         | -                       |

**OSHA** - Occupational Safety and Health Administration Not applicable

**CERCLA** Not applicable

**California Proposition 65** This product does not contain any Proposition 65 chemicals.

**U.S. State Right-to-Know Regulations**

| Component               | Massachusetts | New Jersey | Pennsylvania | Illinois | Rhode Island |
|-------------------------|---------------|------------|--------------|----------|--------------|
| Benzonitrile, 4-fluoro- | -             | X          | X            | X        | X            |

**U.S. Department of Transportation**

Reportable Quantity (RQ): N  
 DOT Marine Pollutant Y  
 DOT Severe Marine Pollutant N

**U.S. Department of Homeland Security** This product does not contain any DHS chemicals.

**Other International Regulations**

**Mexico - Grade** No information available

**Authorisation/Restrictions according to EU REACH** Not applicable

| Component               | CAS No    | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-------------------------|-----------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Benzonitrile, 4-fluoro- | 1194-02-1 | -                                                                   | -                                                                             | -                                                                                                     |

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

| Component               | CAS No    | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|-------------------------|-----------|----------------|------------------------------|---------------------------|--------------------------------------------|
| Benzonitrile, 4-fluoro- | 1194-02-1 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

**Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?**

Not applicable

**Other International Regulations**

| Component | CAS No | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-----------|--------|------------------------------------------------------------------------------|-----------------------------------------------------------------------------|----------------------------|------------------------------------|
|           |        |                                                                              |                                                                             |                            |                                    |

|                         |           | Notification   | Requirements   |                |                |
|-------------------------|-----------|----------------|----------------|----------------|----------------|
| Benzonitrile, 4-fluoro- | 1194-02-1 | Not applicable | Not applicable | Not applicable | Not applicable |

## 16. Other information

**Prepared By** Regulatory Affairs  
Thermo Fisher Scientific  
Email: EMSDS.RA@thermofisher.com

**Revision Date** 01-Sep-2023  
**Print Date** 01-Sep-2023  
**Revision Summary** This document has been updated to comply with the US OSHA HazCom 2012 Standard replacing the current legislation under 29 CFR 1910.1200 to align with the Globally Harmonized System of Classification and Labeling of Chemicals (GHS).

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**